



1. OFDM

◆ Abbreviation

Orthogonal Frequency Division Multiplexing

◆ Definition

👉 A technique that **divides data into many small signals and sends them on different frequencies (subcarriers)**.

◆ Main Points

- Uses **multiple subcarriers**
- Subcarriers are **orthogonal (no interference)**
- Uses **Cyclic Prefix (CP)**
- Simple and widely used
- Medium spectral efficiency

◆ Example

👉 Used in **4G LTE networks**



2. UPMC

◆ Abbreviation

Universal Filtered Multi-Carrier

◆ Definition

👉 An improved OFDM where **groups of subcarriers are filtered** to reduce noise.

◆ Main Points

- Filters **sub-bands (group of carriers)**
- Less noise than OFDM
- No full CP (or very small)
- Better spectral efficiency
- Balanced performance

◆ Example

👉 Used in **5G waveform research**

3. FBMC

◆ Abbreviation

Filter Bank Multi-Carrier

◆ Definition

👉 A technique where **each subcarrier is filtered individually for very clean signals.**

◆ Main Points

- Strong filtering per subcarrier
- No Cyclic Prefix
- Very low interference
- Very high spectral efficiency
- Complex to implement

◆ Example

👉 Used in **advanced 5G systems / research**

4. OFDMA

◆ Abbreviation

Orthogonal Frequency Division Multiple Access

◆ Definition

👉 A technique where **different users use different subcarriers**.

◆ Main Points

- Multi-user version of OFDM
- Each user gets **separate frequency**
- Low interference
- Easy implementation
- Medium capacity

◆ Example

👉 Used in **4G LTE (downlink)**

5. NOMA

◆ Abbreviation

Non-Orthogonal Multiple Access

◆ Definition

👉 A technique where **multiple users share same frequency using different power levels**.

◆ Main Points

- Uses **power domain**
- Users share same channel
- High spectral efficiency
- Supports many users
- Needs **SIC (complex receiver)**

◆ Example

👉 Used in **5G networks**



SUPER EASY FINAL MEMORY



- OFDM → Divide signal
- UFMC → Filter groups
- FBMC → Filter each
- OFDMA → Divide users
- NOMA → Share users